Water Air Soil Pollut (2026) 237:327
https://doi.org/10.1007/s11270-025-08996-8

Life-Cycle Greenhouse Gas Emissions Accounting for Sanitary landfills – a Case Study of a Typical Sanitary Landfill in Hangzhou, China

Shuai Zhang · Kai Liu · Xuyang Huang · He Li · Wenjie Xu · Yuzhong Zhang · Guangyao Li · Liangtong Zhan · Jiwu Lan · Yunmin Chen

Received: 16 January 2025 / Accepted: 6 December 2025
© The Author(s), under exclusive licence to Springer Nature Switzerland AG 2025

Abstract Sanitary landfill is a widely used municipal solid waste (MSW) treatment method worldwide. However, existing research on greenhouse gas (GHG) emissions accounting for sanitary landfills is insufficiently detailed, resulting in lagging development of GHG reduction strategies and measures. This study proposes a comprehensive life-cycle GHG emissions accounting framework that incorporates material transformation laws and energy conversion processes to evaluate the GHG emissions and reductions of sanitary landfills. The potential for achieving carbon neutrality in well-managed sanitary landfills is also explored. Key findings include: (1) Methane from uncollected landfill gas (LFG) is the largest contributor to GHG emissions, accounting for 89.8% of the total emissions. (2) The LFG collection rate is identified as the most critical factor influencing GHG emissions. (3) A well-managed sanitary landfill has the potential to achieve carbon neutrality or even negative. The proposed framework offers an effective approach for accounting GHG emissions and exploring the carbon neutrality potential of sanitary landfills.

Keywords MSW · Sanitary landfill · Life Cycle Assessment · Greenhouse gas · Landfill gas · Carbon neutrality

S. Zhang · K. Liu · X. Huang · H. Li · W. Xu (✉) · L. Zhan · J. Lan · Y. Chen
MOE Key Laboratory of Soft Soils and Geoenvironmental Engineering, Zhejiang University, Hangzhou 310058, China
e-mail: wenjiexu@zju.edu.cn

Y. Zhang
Key Laboratory of Coastal Environment and Resources of Zhejiang Province, School of Engineering, Westlake University, Hangzhou 310030, Zhejiang, China

G. Li
Key Laboratory of Urban Security and Disaster Engineering of Ministry of Education, Beijing University of Technology, Beijing 100124, China

1 Introduction

The greenhouse effect, driven by greenhouse gases (GHG), is a primary cause of global warming (Wang & Geng, 2015). Annually, around 50 billion tonnes of GHG are emitted globally (Ritchie et al., 2020). Within these emissions, the waste sector contributed 3.2% of total GHG emissions in 2016, ranking fourth after energy, agriculture, and industry. While the emissions from the waste sector represent a relatively small percentage of global GHG emissions, taking a systems perspective reveals that addressing waste management upstream can contribute to a significant reduction of 15–20% in global GHG emissions. This can be achieved by focusing on waste prevention, diversion from disposal through recycling and treatment, and utilizing by-products to offset emissions associated with resource extraction or the use of carbon-intensive materials like chemical fertilizers. When it comes

Published online: 23 December 2025

to GHG emission reductions, waste management becomes the second area of study worth exploring, following energy (Xin et al., 2020).

In 2016, China produced 19.1% of the world's municipal solid waste (MSW), the highest globally (World Bank, 2020)(Fig. S1a). This presents a significant opportunity for reducing GHG emissions through proper landfill disposal, aiding in carbon peaking and neutrality goals. Sanitary landfills are the primary MSW disposal method in China, accounting for 33.14% of waste disposal in 2021, despite a declining trend from 2003 to 2020 (Fig. S1b). Globally, 37% of collected waste is landfilled (Kaza et al., 2018). Landfill gas (LFG) from landfills primarily consists of CH_4 and CO_2. CH_4 has a global warming potential 28 times greater than CO_2 over 100 years and 84 times greater over 20 years (Eggleston et al., 2006). In the United States, MSW landfills were the third-largest source of anthropogenic CH_4 emissions in 2016, accounting for 16.8% of the total (EPA 2020). In Western Europe, Japan, and Korea, incineration is common for MSW (Brunner & Rechberger, 2015; De Feo & Malvano, 2009; Pan et al., 2015). However, incineration's environmental benefits are not always superior to those of landfills (Anshassi et al., 2021). Capturing CH_4 and CO_2 from landfills for energy recovery can lead to significant GHG reductions. Net GHG emissions from sanitary landfills can be reduced to zero or even become negative, meaning the GHG offsets from LFG-based electricity generation surpass the emissions, not due to direct CO_2 capture or soil carbon sequestration (Manfredi & Christensen 2009). The environmental benefit potential of sanitary landfills is greater than that of incineration (Pandey et al., 2018, Pietzsch et al., 2017, Samadder et al., 2017, Vaverková, 2019).

Life cycle assessment (LCA) is a valuable tool to assess and quantify the environmental benefits of GHG emissions (Anshassi et al., 2021), which is widely used by researchers in the field of MSW management to evaluate environmental impacts (Friedrich & Trois, 2016; Ghosh et al., 2019; Liu et al., 2017; Turner et al., 2015). However, our proposed framework differs significantly from existing LCA methods in several key aspects: (1) Material transformation focus: Our framework explicitly tracks carbon and nitrogen transformation pathways from MSW through biochemical degradation to final GHG emissions, while conventional LCA typically treats waste decomposition as a black box; (2) Dual-process integration: We simultaneously consider biochemical processes (anaerobic degradation, nitrification/denitrification) and energy conversion processes (fossil fuel combustion, electricity generation), providing more comprehensive emission accounting; (3) Dynamic temporal modeling: Unlike static LCA approaches, our framework captures the temporal dynamics of gas generation using a two-stage model specifically developed for high organic content Chinese MSW; (4) Comprehensive boundary definition: We include often-overlooked emission sources such as N2O from leachate treatment, which can contribute up to 72.8% of total emissions in some cases.

Previous studies have utilized LCA to estimate the carbon footprints of waste incineration and landfill, enabling comparisons of GHG emissions across different waste management systems and optimizing these systems (Arena et al., 2003; Consonni et al., 2005; Zhao et al., 2011). However, it is important to note that the GHG emission estimation for sanitary landfill needs more refinement, and the accounting boundary delineation for sanitary landfill GHG emissions in LCA studies is incomplete enough. In particular, the GHG emissions of N_2O and CH_4 from landfill leachate treatment should be considered. Leachate treatment in landfills is a significant N_2O source, which can contribute 72.8% of total emissions (Wang et al., 2017). These emissions from leachate treatment present an important GHG emission reduction potential compared to LFG emissions.

The large-scale Hangzhou sanitary landfill (HSL) is selected as the case study to: (1) clarity the life-cycle GHG emissions composition and formation mechanism of sanitary landfills based on carbon and nitrogen transformation law and energy conversion; (2) propose a comprehensive life-cycle GHG emission accounting framework for sanitary landfills; (3) quantify the life-cycle GHG emissions of HSL, identify critical factors for emission reductions, provide decision-makers with a reference basis for reducing GHG emissions, and explore the possibility of carbon neutralization in well-managed sanitary landfills.

2 Study Area: Hangzhou Sanitary Landfill (HSL), China

The large-scale Hangzhou sanitary landfill (HSL), located in Qinglongwu Valley, Hangzhou City, Zhejiang Province, China (Fig. 1), consists of two stages.

The first-stage project was China's first standardized valley-type MSW landfill. The second stage, operational from 2007 to 2020, includes features like a comprehensive cover system, LFG treatment system, and leachate treatment system. The landfill accommodates about 4,300 tons of MSW per day, peaking at 8,300 tons in July 2017 (Table S1).Food waste constitutes around 60% of the total weight of the MSW (Zhan et al., 2017). To mitigate environmental impacts, the landfill uses a high-density polyethylene (HDPE) cover. Leachate is collected via drainage layers and treated at a nearby plant. LFG is partly flared and partly used for electricity generation at a biogas power plant. Monitoring wells assess changes in chemical oxygen demand (COD) and total nitrogen (TN) through downstream drilling and groundwater tests.

3 Life-Cycle GHG Emissions Accounting Framework

3.1 Material Transformation Law and Energy Conversion Processes

The GHG from sanitary landfills are formed and emitted with carbon and nitrogen transformation law and energy conversion processes. It is a good way to figure out the composition and formation mechanism of GHG emissions in the perspective of material transformation law and energy conversion processes. As shown in Fig. S2, The life span can be divided into four stages with clearly defined system boundaries: (1) Planning and construction stage: includes GHG emissions from raw material extraction, transportation, and on-site construction activities, facility manufacturing, and infrastructure development; (2) Transportation stage: covers direct fuel combustion emissions from waste collection and transportation vehicles, including both collection routes and transfer operations; (3) Landfilling stage: encompasses waste decomposition emissions, energy consumption for operations, leachate treatment, and LFG collection/utilization systems; (4) Post-closure stage: includes long-term monitoring, maintenance activities, and continued LFG generation/collection. The GHG emitted from sanitary landfills mainly are CH_4, CO_2, and N_2O. Tracking back to the formation mechanism, the GHG in sanitary landfills come from carbon and nitrogen in MSW.

The carbon and nitrogen in MSW exist in different forms during the life span and emitted as GHG finally (Fig. 2.a). Before landfilling, carbon and nitrogen in MSW mainly exist in organic degradable waste, which undergoes biochemical degradation

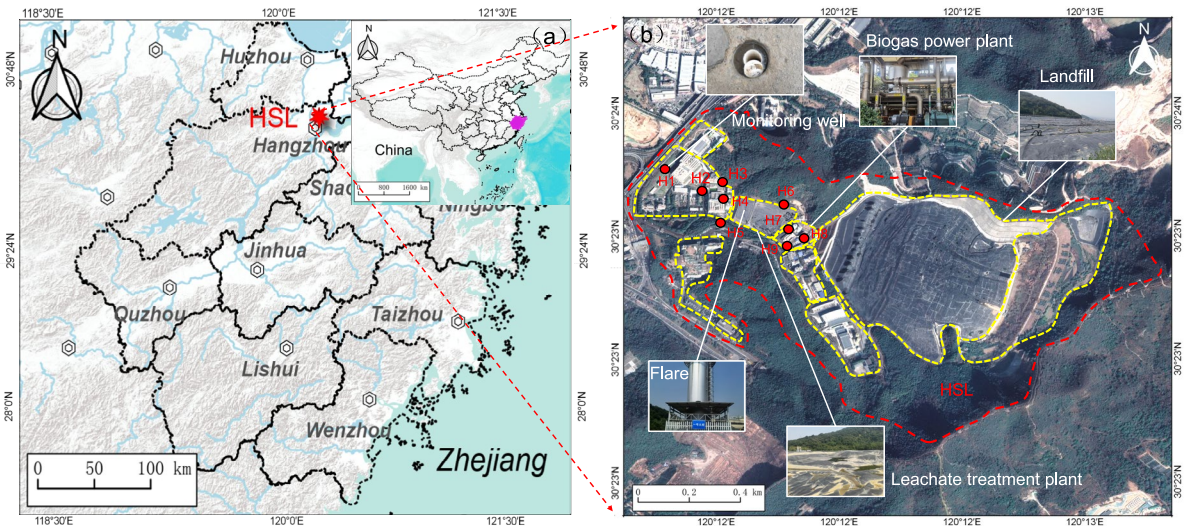

Fig. 1 Location and overview of the HSL: **a**) HSL is located in Hangzhou, Zhejiang Province, China. **b**) Surrounding environment and facilities distribution of HSL. MSW is dumped in the landfill area. Part of generated LFG is flared for power generation in the biogas power plant. The remaining generated LFG is flared into the atmosphere. The generated leachate is collected to the leachate treatment plant. Monitoring wells H1-9 are drilled for groundwater tests

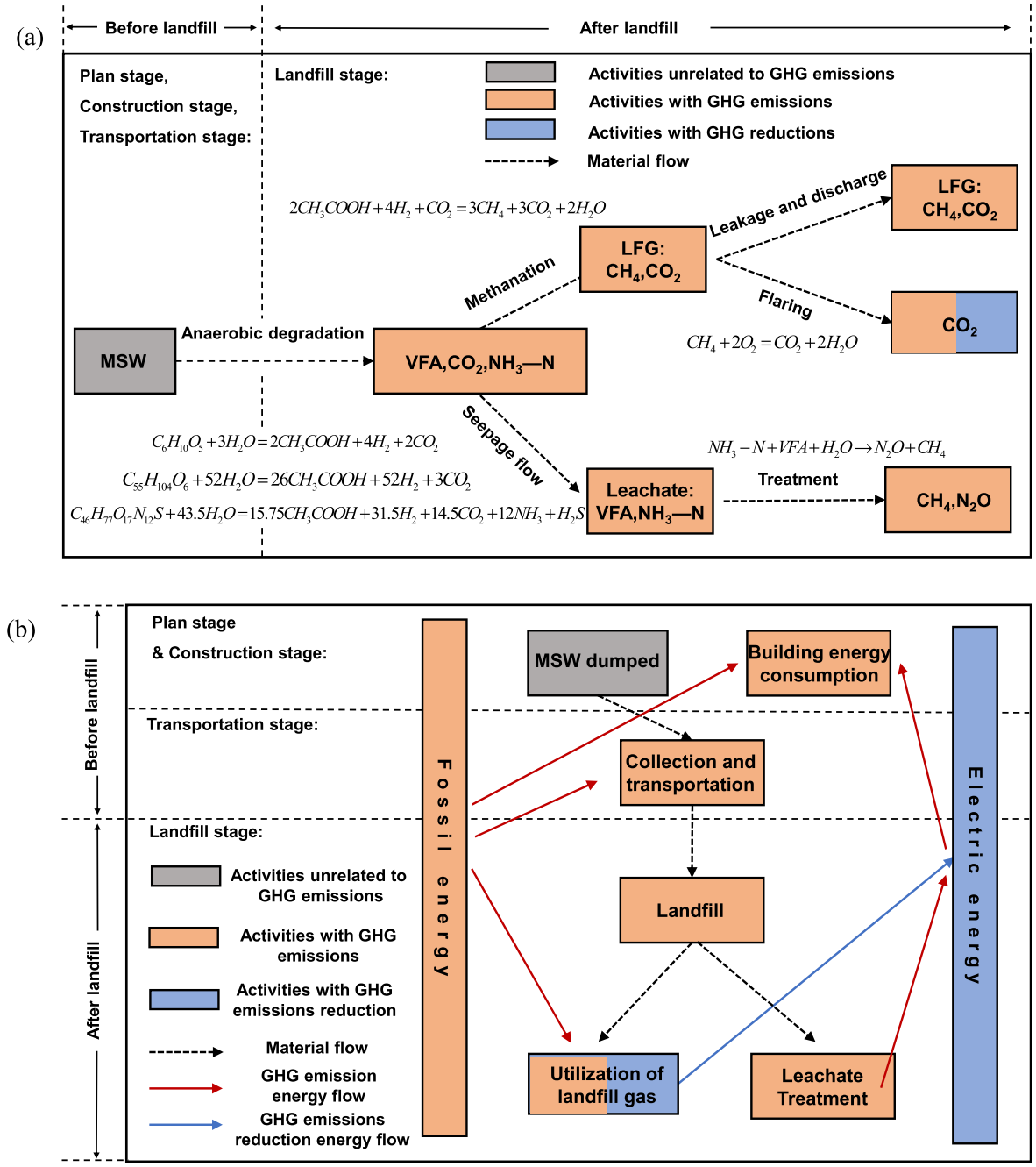

Fig. 2 Accounting principles: **a**) Material transformation laws: life-cycle carbon and nitrogen transformation of MSW in sanitary landfills. **b**) Energy conversion processes: life-cycle energy conversion of MSW in sanitary landfills

in the landfill. The main organic degradable waste components, represented by sugars and fats, are hydrolysed to volatile fatty acid (VFA) and CO_2 in the anaerobic environment of the sanitary landfills (Zhan et al., 2017). In the subsequent anaerobic degradation phase, VFA combines with H_2 and CO_2 to form CH_4 and CO_2, eventually forming the LFG with CH_4 and CO_2 as the main components. A portion

of LFG can be emitted directly into the atmosphere due to leaks from the landfill or human-initiated discharge. Another part of LFG is collected by pipeline and transported to flare combustion or biogas power plants to be burned for electricity generation, eventually emitted to the atmosphere as CO_2. The degradable organic waste in landfills also produces ammonia nitrogen during anaerobic degradation. Ammonia nitrogen and a portion of VFA are fused with leachate under seepage flow into the leachate treatment plant, where they are released to the atmosphere as CH_4 and N_2. During the processes of nitrification and denitrification leachate treatment, N_2O, as an intermediate product, can be released into the atmosphere during aeration. (Eggleston et al., 2006).

Besides material transformation, the GHG emitted in sanitary landfills also occur in energy conversion processes. As shown in Fig. 2.b, the GHG emissions mainly come from: (1) Fossil fuel combustion provides energy for landfill construction, waste transportation vehicles, and grid power generation; (2) The power grid provides electrical power for building operations and operational equipment in the landfill, especially in the leachate treatment plant; (3) LFG, transported to the biogas power plant, will be used for combustion to generate electricity, reducing the landfill's dependence on grid power and contributing to GHG emission reductions. Combining the material transformation and energy conversion processes, the life-cycle GHG emissions in sanitary landfills can be concluded.

3.2 Four GHG Emission Sources

Based on the analysis of the material transformation law and energy conversion processes, the sources and calculation objects of life-cycle GHG emissions are figured out. As shown in Fig. S2, the life-cycle GHG emissions of sanitary landfills mainly come from four parts: (1) landfill buildings, GHG emissions from landfill buildings come from the consumption of construction materials during construction and electricity consumption for daily operation and maintenance of landfill buildings; (2) MSW collection and transportation, the waste hauler collects the MSW from the residents and transports it to sanitary landfills, during which the waste hauler consumes fuel and emits CH_4, CO_2, and N_2O, which are the primary sources of GHG emissions; (3) LFG utilization, LFG utilization focuses on the contribution of uncollected LFG emissions GHG emissions and GHG emission reductions from biogas power plants for electricity generation; (4) leachate treatment, N_2O and CH_4 emitted after leachate treatment are the primary sources of GHG emissions.

3.3 Life-Cycle GHG Emissions Accounting

3.3.1 Scope Definition

This part is used to define the life-cycle GHG emissions accounting boundary for sanitary landfills. The accounting boundary defines the data calculation scope of the four parts of GHG emissions accounting.

Geographical boundary: The boundary for MSW collection and transportation is defined by the landfill's coverage within the city, determined by the MSW quantity and transport routes. GHG emissions from collection and transportation are limited to the city, while emissions from landfill buildings, LFG utilization, and leachate treatment are confined to the landfill site, which serves as the geographic boundary for these emissions (Fig. 1.b).

Time boundary: The second-stage project in HSL entered ecological restoration after closing in 2020, with relevant facilities abandoned. These buildings can be dismantled and reused. The timeframe selection is based on multiple factors: (1) 2007 marks the commencement of HSL's second-stage operations with systematic data collection; (2) 2056 represents the end of the designed operational lifespan (50 years) plus post-closure monitoring period as per Chinese regulations; (3) Continuous monitoring data from 2007–2022 provides a solid foundation for model validation; (4) This timeframe aligns with China's carbon neutrality timeline and waste management policy considerations.

3.3.2 GHG Emissions of Sanitary Landfill Buildings

The GHG emissions associated with sanitary landfill buildings can be categorized into two components: the GHG emissions from construction materials and the GHG emissions resulting from electricity consumption during the daily operation and maintenance of the buildings.

GHG emissions from construction materials contribute approximately 9.65% to 29.28% of the total

life-cycle emissions of buildings (Tae et al., 2011; Zhang & Wang, 2015). Calculating these emissions involves extensive data collection, which can be challenging. However, a model based on building floor area was selected because all main structures (e.g., offices, leachate treatment units) at the landfill use reinforced concrete construction. This allows for consistent estimation of GHG emissions using building area as a proxy, aligning with established practices in Chinese infrastructure LCA (Zhang & Wang, 2015). Both sanitary landfill and residential buildings in the model typically have reinforced concrete structures. Using floor area data, Eq. (1) estimates the GHG emissions per unit area of construction materials in sanitary landfills.

The GHG emissions of the electrical energy consumption during the daily operation and maintenance of sanitary landfill buildings, both during operation and closure, were determined using bottom-up approach. The electrical energy consumption is estimated by the category of buildings in the sanitary landfill during 2007–2056 (Table 1). Buildings in sanitary landfills can be divided into small and medium-sized public buildings and large public buildings by the standard of the floor area standard of 20,000 m^2. The average annual energy consumption for large public buildings is 113.9 kWh/m^2, while small and medium-sized public buildings have an average annual energy consumption of 82.53 kWh/m^2. By statistically analyzing the landfill building area, the average annual electrical energy consumption of the corresponding buildings can be estimated. To estimate the total electrical energy consumption during the period from 2007 to 2056, the operational lifespan of the buildings is taken into account. The total electrical energy consumption is then multiplied by the electricity GHG emission factor to derive the GHG emissions associated with the electrical energy consumption of the landfill buildings. The estimation of GHG emissions for sanitary landfill buildings can be conducted using Eq. (2).

$$LCCO_{2\cdot MAT} = 297.490 + 0.007 \times A \quad (1)$$

$$E_{building} = LCCO_{2\cdot MAT} \cdot A + e_{building} \cdot K_e/1000 \quad (2)$$

where:

$LCCO_{2\cdot MAT}$	GHG emissions per unit area of building materials (t/m^2);
A	total area of a single building (m^2);
$E_{building}$	GHG emissions of sanitary landfill buildings (t);
$e_{building}$	electricity consumption during the daily operation and maintenance ($kW \cdot h$);
K_e	electricity GHG emission factor, $0.7921(kg/(kW \cdot h))$;

The factor 1000 is unit conversion between tons and kilograms.

3.3.3 GHG Emissions of MSW Collection and Transportation

The GHG emissions associated with the collection and transportation of MSW primarily stem from the fossil energy consumption of the vehicles utilized for waste hauling. The waste haulers entail diesel engines with a 5-ton capacity. To estimate the diesel consumption and GHG emissions of these waste haulers, emission factors are used for a 5-ton vehicle, where the coefficients for diesel consumption, CO_2

Table 1 Physical components of MSW in HSL

Parameters	Catering waste	Paper	Textile	Bamboo
Proportion (%)	53.96	18.67	17.64	3.08
DOC (%)	32.41	38.78	47.63	42.93
Water content (%)	84.3	25.5	29.7	36.5
Rapidly degrading substances (%)	85	47.4	15	15
Slowly degrading substances (%)	15	52.6	85	85
Wet basis(2009) (%)	52.9	12.05	3.21	0.89
Wet basis(2010) (%)	56.8	9.31	3.05	1.92
	56.23	15.65	2.01	1.55
Average Wet basis (%)	55.31	12.34	2.76	1.45

emissions, CH_4 emissions, and N_2O emissions are estimated at 0.238 $kg/(ton \cdot km)$, 0.75 $kg/(ton \cdot km)$, 4.22E–5 $kg/(ton \cdot km)$, and 1.92E–5 $kg/(ton \cdot km)$ (Zhao et al., 2011), respectively. By considering the transport distance of the waste hauling vehicles, the diesel consumption and corresponding GHG emissions can be estimated in accordance with these coefficients.

In this study, it is assumed that waste logistics networks in large Chinese cities are similarly structured. Jinan's MSW logistics network is analyzed as a model. Factors considered for estimating daily transport distance include population, urbanization rate, consumption level, daily waste generation, city area, and number of transfer stations. A linear regression model estimates the daily transport distance for MSW collection and transportation based on daily waste quantity $\boldsymbol{M}$ and city area $\boldsymbol{S}$ (Eq. (3)). GHG emissions from collection and transportation are estimated using Eq. (4). Applying the model to Chengdu, with similar urbanization and waste generation rates, yields an estimated average daily transport distance of 49.435 km for Jinniu district, closely matching the actual 53.83 km with an 8.9% deviation.

For HSL, the average daily waste collection quantity $\boldsymbol{M}$ can be calculated from the annual waste landfill quantity between 2007 and 2020, as shown in Table S1. The transport distance can be estimated using Eq. (3) based on the calculated waste collection quantity and the area of Hangzhou city. The GHG emissions associated with MSW collection and transportation can then be calculated using Eq. (4), with coefficients from National III waste hauler and the estimated transport distance.

$$d = 1.858S^{1.063} + 3.646M^{0.9981} + 0.00003856 \quad (3)$$

$$E_{collection} = 0.238 \cdot 5d \cdot (0.75 + 28 \cdot 4.22 \times 10^{-5} + 265 \times 1.92 \times 10^{-5}) \quad (4)$$

where:

d	transport distance (m);
S	city region area (m^2);
M	daily waste collection quantity (t);
$E_{collection}$	GHG emissions of MSW collection and transportation;

The factor 28 is the 100-year warming potential of CH_4;

The factor 265 is the 100-year warming potential of N_2O.

3.3.4 GHG Emissions of LFG Utilization

The GHG emissions associated with LFG utilization come from CH4 in uncollected LFG, while GHG emission reductions occur through electricity generation at biogas power plants.

The Landfill Gas Emissions Model (Land GEM),which is recommended by EPA, is specifically used to calculate the LFG emissions from landfills (Atabi et al., 2014; Fallahizadeh et al., 2019). Unlike MSW composition in the United States, China's MSW contains a significant proportion of kitchen waste (Zhan et al., 2017). As a result, the decomposition of Chinese MSW enters the methanogenesis phase rapidly after disposal, leading to substantial LFG production within a short period. The LFG production rate exhibits a distinct pattern with a large and swift peak that is quickly achieved. Once the degradation of kitchen waste and other components is mostly complete, the LFG production rate experiences a rapid decline and tends to level off. This process showcases distinct stages and a noticeable stepwise pattern. The Land GEM is not suitable for calculating LFG production from Chinese MSW, considering the high proportion of kitchen waste and its distinct characteristics of staged gas production rates. In response to these characteristics, a two-stage model has been proposed to more accurately estimate LFG production from Chinese MSW (Eq. (5)). Equation (5) is based on the two-stage gas production model specifically for Chinese MSW with high organic content. The model accounts for the distinct decomposition phases: rapid degradation of easily biodegradable materials (food waste, paper) and slow degradation of lignocellulosic materials (wood, textiles). This approach better represents the biochemical reality of anaerobic digestion in Chinese landfills compared to the single-phase first-order decay models (LandGEM) developed for Western MSW compositions.

$$Q_n = \sum_{i=1}^{n} M_i L_{R0} \left(\frac{k_R}{12}\right) e^{-\frac{k_R}{12}(t_i - t_{R,lag})} + \sum_{i=1}^{n} M_i L_{S0} \left(\frac{k_S}{12}\right) e^{-\frac{k_S}{12}(t_i - t_{S,lag})} \tag{5}$$

where:

Q_n the theoretical gas production rate per unit time of LFG in the ith month after the commencement of landfill operation (m^3/month);

t_i the landfill age of the ith month of the landfilled MSW at the calculation time;

M_i the quantities of MSW landfilled in the ith month (t);

L_{R0} the theoretical gas production rate of rapidly degrading substances per unit mass of waste (m^3/t);

L_{S0} the theoretical gas production rate of slowly degrading substances per unit mass of waste (m^3/t);

k_R the gas production rate constant for rapidly degrading substances (yr^{-1});

k_S the gas production rate constant for slowly degrading substances (yr^{-1});

$t_{R,lag}$ the gas production lag time for rapidly degrading substances, 6 months;

$t_{S,lag}$ the gas production lag time for slowly degrading substances, 12 months.

The parameters in Eq. (5) are determined as follows: L_0 values are determined through laboratory analysis of HSL waste samples collected during 2009–2011; k values are derived from field pumping experiments achieving steady-state conditions; lag times are based on Chinese MSW decomposition characteristics with high organic content.

Table S1 presents the monthly and yearly quantities of MSW landfilled at HSL. From 2009 to 2011, samples from drilling experiments were analyzed to determine the percentages of dissolved organic carbon (DOC), water content, and wet basis ratios of four types of degradable organic wastes: food waste, paper, textile, and bamboo materials (Table 1). According to the experimental analysis results, L_{R0} is determined to be 115.52 m^3/Mg, while L_{S0} is calculated to be 13.36 m^3/Mg (Eqs. (6 ~ 8)).

$$L_0 = 1867 \times \sum_{1}^{4} [W_i \times (1 - d_i) \times DOC_i] \tag{6}$$

$$L_{R0} = L_0 \cdot \frac{\sum_i W_i \cdot (1 - d_i) \cdot \varphi_i^R}{\sum_i W_i \cdot (1 - d_i) \cdot \varphi_i^R + \sum_i W_i \cdot (1 - d_i) \cdot \varphi_i^S} \tag{7}$$

$$L_{S0} = L_0 - L_{R0} \tag{8}$$

where:

L_0 maximum CH_4 gas production potential ($m^3 CH_4 Mg_{waste}{}^{-1}$); the factor 1867 is the conversion factor between the unit mass of organic carbon and volume of LFG in standard conditions (m^3/Mg);

DOC_i the content of degradable organic carbon of the ith fraction (dry basis, %);

W_i wet weight content of the ith component (wet basis, %);

d_i water content of the ith component (wet basis, %);

φ_i^R dry basis content of rapidly degrading substances;

φ_i^S dry basis content of slowly degrading substances.

The gas production rate is determined through field pumping experiments. When the pumping operation reaches a steady state, it can be assumed that all the LFG produced within the pumping influence area is effectively pumped out. The LFG production within the radius of influence can be considered equal to the collection volume at the steady state, which is used to calculate the LFG production rate (Eq. (9)). The experimental results for gas production rate are presented in Table 2, a value of 0.8 (yr^{-1}) is used for k_R, while the value of 0.2 (yr^{-1}) is taken as the value for k_S. The LFG production of two-stage model can be calculated based on the Eq. (5).

$$Q_t = \sum_i \pi R^2 h_t \rho \left[L_{R0} k_R e^{-\frac{k_R}{12}(t-6)} + L_{S0} k_S e^{-\frac{k_S}{12}(t-12)} \right] \quad (9)$$

where:

Q_t	the volume of total gas collected (m^3);
R	pumping influence radius (m);
h_t	pumping well depth (m);
ρ	the density of MSW (kg/m^3);
t	the duration of pumping

The HSL is a well-managed sanitary landfill that includes a comprehensive gas recovery and utilization system. The utilization of LFG at HSL from 2010 to 2021 is depicted in Fig. S3. In order to assess the overall GHG emission potential of LFG, the LFG production estimated by the two-stage model is used as a comparative measure. With the continuous improvement of the HSL LFG recovery system, the amount of collected LFG has been steadily increasing. In 2021, the collected amount exceeded the estimated production of the two-stage model, resulting in a LFG collection rate of over 100%. This may be due to the accumulation of previously generated LFG remaining insufficiently extracted from the landfill. In contrast, the utilization of LFG for electricity generation remains relatively unchanged each year due to limited LFG utilization facilities (Table 3). As the amount of collected LFG increases, the utilization rate of LFG decreases and reached its lowest point at 19.73% in 2021.

The GHG emissions associated with LFG are calculated using Eq. (10). The GHG emissions come from CH_4 in uncollected LFG. Utilizing LFG for electricity generation reduces GHG emissions, but this is limited by the volume of LFG and the capacity of generation units. GHG emission reductions are calculated with Eq. (11). After the landfill closure in 2021, 620,000 m^3 of LFG is collected daily, generating 370,000 kWh of electricity per day. Considering LFG collection rates from 2010 to 2021 (Fig. S3) and HSL's operational conditions, a 30% collection rate is assumed before 2020, increasing to 90% after closure in 2021 (Tables 4 and 5).

$$E_{LFG} = 28 \times \frac{16}{2} \times \frac{V_u}{24.5} \quad (10)$$

$$E_e = e \cdot K_e \quad (11)$$

where:

E_{LFG}	LFG GHG emissions (t);
E_e	GHG emission reductions from LFG combustion for electricity generation (t);
V_u	the volume of direct LFG emissions (m^3); the factor 24.5 is the molar volume of gas at room temperature; the factor 28 is CH_4 warming potential;
e	LFG combustion for electricity generation ($kW \cdot h$);
K_e	electricity GHG emission factors, 0.7921 $kg/(kW \cdot h)$.

3.3.5 GHG Emissions of Leachate Ttreatment

The GHG emissions associated with leachate treatment are attributed to CH_4 and N_2O emissions resulting from the VFA and total nitrogen (TN) in the

Table 2 LFG collection rate in pumping experiments

Leachate water level	Pumping pressure (kPa)	Influenced radius (m)	LFG collection rate (m^3/h)
Initial leachate water level	−1	25	55.9
Lowered leachate water level	−1	28	55.3

Table 3 Changes of power generation facilities in LFG power plant

Time Power generation facilities	2007–2010	2011–2015	2016–2019	2019-after
Number of generator sets	2	8	12	16
Installed capacity of the generators($kW \cdot h$)	1.94	8.24	12.24	16.36

leachate. Monthly leachate volume in the HSL during 2009–2056 is shown in Table 6 based on experimental monitoring data and model prediction (He & Hu, 2022). The distribution of COD and TN in leachate samples obtained from wells H1-4 (Fig. 1b) is illustrated in Figs. S5a & S5b. The COD concentrations range from 5000 to 30,000 mg/L, while the TN concentrations range from 2500 to 6500 mg/L. For the purposes of determining GHG emissions from leachate treatment, COD and TN values of 20,000 mg/L and 5000 mg/L, respectively, are used as the basis for calculation, specifically in the most densely distributed location. Using Eqs. (12 ~ 16) from Chapter 6 of IPCC (Eggleston et al., 2006), the GHG emissions resulting from CH_4 and N_2O to leachate treatment can be calculated as follows:

$$E_{leachate} = 28 \cdot CH_4 emission + 265 \cdot N_2 Oemission \tag{12}$$

$$CH_4 emission = \sum_i \left[\left(TOW_i - S_i \right) EF_i - R_i \right] \tag{13}$$

$$TOW_i = V \cdot COD_i \tag{14}$$

Table 4 GHG emissions of HSL landfill buildings during 2007–2056

Buildings	Floor area (m^2)	GHG emissions of construction materials (t)	Annual GHG emissions of electric energy (t)	Total GHG emissions (t)
No.1 leachate treatment plant	9,760	3,570.31	566.66	31,903.51
No.2 leachate treatment plant	9,957	3,656.10	578.10	32,561.20
Water resources recycling center	26,038	12,491.89	1,511.76	88,080.01
Biogas power plant	10,883	4,066.66	631.87	35,659.93
No.1 catering waste treatment plant	8,667	3,104.16	503.20	28,264.40
No.2 catering waste treatment plant	54,629	37,141.87	4,377.35	256,009.28
No.1 household kitchen waste treatment plant	6,000	2,036.94	348.36	19,454.90
Classification and reduction of center	53,404	35,851.07	4,279.19	24,9810.59
Park parking and security building	43,193	25,908.93	3,461.00	198,958.77
Total	222,531	127,827.93	16,257.49	940,702.59

Table 5 GHG emissions of MSW collection and transportation

Year	Annual transport distance (km)	Diesel consumption (t)	CO_2 emissions (t)	CH_4 emissions (kg)	N_2O emissions (kg)	GHG emissions (t)
2007	24,962,287	29,705	93,609	5,267	2,396	94,391
2008	25,415,209	30,244	95,307	5,363	2,440	96,104
2009	24,884,364	29,612	93,316	5,251	2,389	94,096
2010	24,954,158	29,695	93,578	5,265	2,396	94,360
2011	25,093,800	29,862	94,102	5,295	2,409	94,888
2012	25,428,984	30,260	95,359	5,366	2,441	96,156
2013	26,236,055	31,221	98,385	5,536	2,518	99,208
2014	26,995,946	32,125	101,235	5,696	2,592	102,081
2015	27,210,151	32,380	102,038	5,741	2,612	102,891
2016	28,469,075	33,878	106,759	6,006	2,733	107,651
2017	29,805,838	35,469	111,772	6,289	2,861	112,706
2018	27,048,542	32,188	101,432	5,707	2,597	102,280
2019	26,280,095	31,273	98,550	5,545	2,523	99,374
2020	26,190,061	31,166	98,213	5,526	2,514	99,034

$$EF_i = B_o \cdot MCF \tag{15}$$

$$N_2Oemission = N \cdot EF_j \cdot 44/28 \tag{16}$$

where:

$CH_4emission$	CH_4 emissions from leachate treatment (kg/yr);
TOW_i	total degradable organic materials in part i leachate treatment (kg COD/yr);
S_i	organic component removed as sludge in inventory year (kg COD/yr);
i	leachate treatment components;
EF_i	CH_4 emission factor for part i (kg CH_4/kg COD);
R_i	CH_4 recovered from i part leachate treatment (kg CH_4/yr);
V	leachate yield (m^3);
COD_i	is chemical oxygen demand (kg COD/m^3);
B_o	maximum CH_4 production capacity (g CH_4/kg COD), taking the default value of 0.25;
MCF	CH_4 correction factor, taking the value of 0.3 according to the landfill leachate treatment equipment and method;
N	total nitrogen in leachate (kg/m^3);
EF_j	GHG emission factors for N_2O emissions (kg N_2O/kg N), taking the default value of 0.005; the factor 44/28 is the conversion of kg N_2O-N into kg N_2O.

4 Results of GHG Emissions

4.1 GHG Emissions Inventory of the HSL

4.1.1 GHG Emissions of Sanitary Landfill Buildings

The cumulative GHG emissions of landfill buildings during 2007–2056 are 0.94 million tons (Table 4). Of which, GHG emissions from construction materials is 0.127 million tons, accounting for 13.59%. From the life-cycle perspective, GHG emissions from electrical energy consumption are the main source of GHG emissions of landfill buildings.

Table 6 Monthly leachate volume in the HSL during 2009–2056

Month Year	1	2	3	4	5	6	7	8	9	10	11	12	Total (m^3)
2009	0	0	0	0	0	0	0	0	0	0	2,791	1,894	142,444
2010	1,390	2,300	3,788	2,310	2,004	2,684	3,542	2,881	3,281	2,696	1,397	1,774	914,885
2011	1,946	1,097	1,752	1,795	2,079	6,505	2,627	3,459	2,068	1,889	1,835	1,662	873,400
2012	2,598	2,256	3,399	2,488	3,077	4,268	2,185	3,834	2,418	1,830	2,393	2,452	1,010,803
2013	1,951	2,167	2,590	2,602	2,717	4,798	1,910	3,870	2,311	4,857	1,919	2,529	1,042,720
2014	2,087	2,744	2,730	2,781	3,544	3,799	3,977	4,057	3,534	2,231	2,455	1,700	1,084,008
2015	2,607	2,870	3,388	3,925	3,092	5,420	5,623	4,972	3,253	2,629	4,103	2,984	1,363,985
2016	3,592	2,005	2,978	4,838	5,010	5,385	3,572	4,231	4,768	3,980	3,373	2,838	1,419,291
2017	3,343	2,314	4,559	4,122	4,031	3,246	4,230	4,642	4,553	3,833	3,931	2,587	1,384,327
2018	3,666	2,251	2,970	3,290	3,947	5,316	4,056	5,896	4,635	3,160	3,385	3,478	1,404,111
2019	3,055	3,965	3,020	2,781	3,212	4,088	4,245	5,872	3,083	2,285	2,058	2,822	1,231,161
2020	4,258	2,174	3,268	2,347	3,820	6,454	4,818	3,100	4,366	2,444	2,096	1,557	1,239,977
2021	1,142	1,782	2,834	1,474	2,758	3,329	2,308	2,924	2,217	1,723	1,648	1,488	780,423
2022	1,464	1,800	1,947	1,881	1,817	3,610	2,096	2,728	2,032	1,547	1,479	1,325	721,104
2023	1,304	1,642	1,793	1,732	1,674	3,472	1,964	2,600	1,908	1,429	1,365	1,216	671,666
2024	1,199	1,525	1,695	1,637	1,583	3,385	1,882	2,523	1,835	1,360	1,299	1,152	640,594
2025–2056	1,322	1,656	1,812	1,750	1,691	3,489	1,981	2,617	1,925	1,445	1,381	1,231	677,788

4.1.2 GHG Emissions of MSW Collection and Transportation

A total of 1.40 million tons of GHGs were estimated during the process of MSW collection and transportation, 48.5% higher than landfill construction, as listed in Table 5. GHG emissions of MSW collection and transportation are mainly influenced by the amount of MSW landfilled and the transport distance.

4.1.3 GHG Emissions of LFG Utilization

The annual GHG emissions and GHG emission reductions of LFG are shown in Fig. S4. Due to the high proportion of food waste in the MSW at HSL, the peak in LFG generation and GHG emissions occurred in 2018, prior to the cessation of landfilling activities at HSL in 2020. The closure of HSL is an essential point when the GHG emissions of LFG decreased significantly, essentially an increase in LFG collection rate. During 2007–2056, the GHG emissions of LFG utilization are 28.62 million tons, while the GHG emission reductions are 0.733 million tons, 2.56% of the former.

4.1.4 GHG Emissions of Leachate Treatment

The GHG emissions of leachate treatment are 0.924 million tons during 2007–2056. Fig. S6 shows that the GHG emissions of CH_4 from leachate treatment are nearly double the emissions from N_2O. The N_2O produced during leachate treatment is not the primary source of GHG emissions.

4.1.5 Inventory of GHG Emissions

During 2007–2056, the GHG emissions of the HSL mainly come from LFG utilization, which accounts for 89.8%, and the trend of GHG emissions is consistent with that of LFG GHG emissions. Table 7 shows GHG emissions and emission reductions of each part in the life-cycle of the HSL during 2007–2056. GHG emissions from landfill buildings, collection and transportation and leachate treatment account for 2.95%, 4.35% and 2.9% of total GHG emissions, respectively (Fig. 3).

4.2 Verification of the Conservation of LFG

4.2.1 Verification of LFG Production

In addition to the two-stage production model (Chen et al., 2016), the IPCC 2019 and the Land GEM are used to estimate the LFG production from 2007 to 2041. As shown in Fig. 4, the estimated LFG production of the two-stage model is high in the early stage (2007–2020) and lower in the late stage, with an earlier peak. The Land GEM model shows a plateau after peaking, and the IPCC 2019 model's production level falls between the Land GEM and two-stage models after 2019. All three models show good consistency, peaking around 2018–2019 at approximately 450 million cubic meters. Due to the high kitchen waste content in Chinese MSW, the two-stage model more accurately describes LFG production changes and GHG emissions in HSL.

4.2.2 Verification of LFG Emissions

The LFG emissions from various sources result in different GHG emissions. It is necessary to verify the LFG emissions to ensure the accuracy of GHG emission accounting. The landfill working area is the primary source of fugitive LFG leakage. Using static chamber experiments conducted during the landfill period, CH_4 emission fluxes of 129.6 $g/(m^2 \cdot h)$ and CO_2 emission fluxes of 3346.24 $g/(m^2 \cdot h)$ were measured**.** Figs. S7 & 5a present the changes at the working area, temporary cover area, and final cover area of the HSL from 2007 to 2021, as determined by remote sensing interpretation. These changes are summarized in Table S2. The changes in the working area and LFG flux were utilized to estimate the volume of LFG escaping from the working area. The Fig. 5b indicates that the volume of LFG escaping from the working area is even more significant than the volume of LFG collected and utilized during the early stage when the LFG collection rate was low.

There still exists a discrepancy between the sum of various LFG emissions and LFG production, which may be attributed to several factors: (1) a portion of the LFG is retained within the waste piles; (2) the landfill operation area and emission flux level are subject to temporal variations, making the estimation

Table 7 Life-cycle GHG emissions and GHG emission reductions of HSL

Emission sources Emission category	Sanitary landfill buildings	Collection and transportation	LFG utilization	Leachate treatment
GHG emissions (t)	940,703	1,396,157	28,617,303	923,699
GHG emissions reduction (t)	0	0	733,223	0

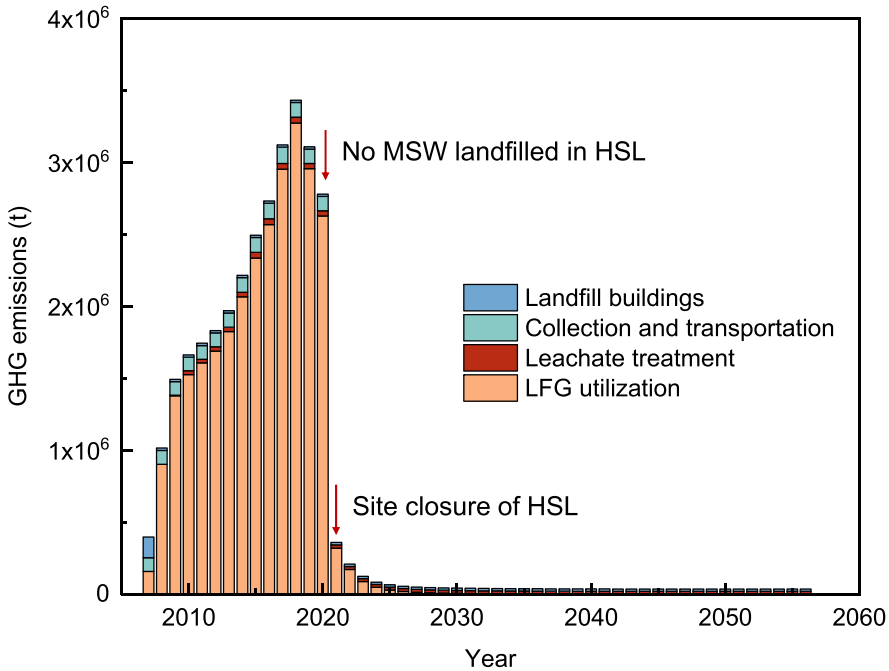

Fig. 3 Life-cycle GHG emissions of HSL during 2007–2056

process challenging by using an overall area average; (3) some LFG does not burn efficiently while passing through the flare. Instead, part of it escapes directly. During the landfill closure stage in 2020, the LFG collection and utilization level increased by 83.27%, which was consistent with the sum of LFG fugacity and production. Moreover, when compared horizontally, the production of CH_4 from leachate treatment was much smaller than LFG production.

5 Discussion: Can a Well-Managed Landfill be Carbon Neutral?

Considering the long-term benefits of landfilling, it is imperative to explore the possibility of achieving carbon neutrality in a well-managed sanitary landfill. The goal of carbon neutrality requires GHG emissions reach net-zero and probably negative somewhere before 2060 (Dong et al., 2022). In sanitary landfills, the anthropogenic GHG emissions mainly come from CH_4 in uncollected LFG, which account for 89.8%. The collected LFG is used for flaring and electricity generation without GHG emissions, resulting in additional GHG reductions instead if the collected LFG are used for electricity generation. The key to carbon neutrality in landfills is the rational use of CH_4 in LFG.

The LFG collection rate is crucial for effective CH_4 utilization. This study analyses the impact of different LFG collection rates (50%, 90%, and 100%) on GHG emissions, using cumulative net GHG emissions as the metric. Achieving "net zero" by 2060 would indicate carbon neutrality. Figure 6 shows that as the LFG

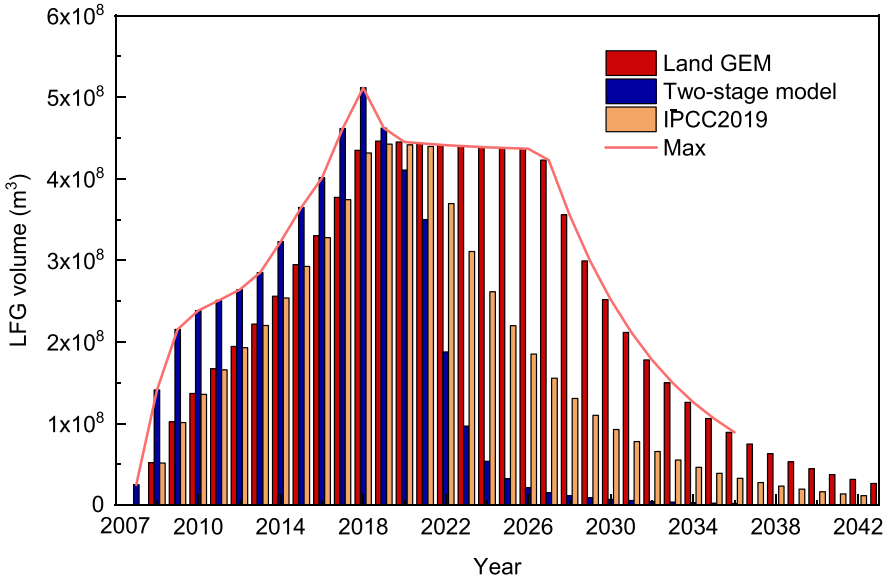

Fig. 4 LFG production of Land GEM, two-stage model and IPCC 2019 during 2007–2042

collection rate increases, cumulative net GHG emissions decrease significantly. At a 100% collection rate, cumulative net GHG emissions drop to 283,438 tons by 2056, just 8.9% of the baseline scenario, nearing net-zero emissions. Notably, from 2018 to 2043, the cumulative net GHG emissions are negative under the 100% collection rate, indicating early GHG reductions. The negative emissions phase (2018–2043) under 100% collection rate results from: (1) Energy displacement: LFG-derived electricity (370,000 kWh/day) displaces grid power with emission factor 0.7921 kg CO_2/kWh, creating emission credits of 107,000 tons CO_2-eq/year; (2) Reduced methane leakage: 100% collection eliminates the 89.8% of total emissions from uncollected LFG; (3) Carbon sequestration: Long-term storage in non-degradable waste fractions contributes additional emission reductions. The combination of these factors results in net emission reductions exceeding remaining operational emissions, creating the negative emissions period. Thus, controlling the LFG collection rate is essential for achieving carbon neutrality in sanitary landfills.

The HSL case study demonstrates significant potential for LFG management improvement within the Chinese context. With the increase in collection rate from 30% (pre-2020) to 90% (post-closure), the facility achieved an 83.27% increase in LFG collection and utilization. This improvement pattern suggests that systematic enhancement of collection infrastructure and operational practices could yield substantial emission reductions across China's landfill sector.

Significant amounts of CO_2 are also emitted from sanitary landfills. Research on Carbon Capture, Utilization, and Storage (CCUS) from industrial facilities suggests potential for additional GHG reductions (Bataille et al., 2018). Technological advancements could further improve LFG electricity generation efficiency, enhancing GHG reductions. Therefore, achieving carbon neutrality is feasible for well-managed sanitary landfills with high LFG collection rates. To achieve this potential, the following policy recommendations are proposed: (1) Regulatory measures: mandatory LFG collection efficiency standards (> 75% as per EU benchmarks), financial incentives for high-efficiency LFG capture systems, and carbon credit mechanisms for verified emission reductions; (2) Addressing barriers: economic analysis shows payback periods for advanced LFG systems range from 2–5 years, technical challenges in older landfills require retrofit solutions, and financing mechanisms through green bonds and government subsidies can overcome initial investment barriers. For landfills receiving substantial organic waste, it is crucial to enhance LFG collection and utilization early to minimize CH_4 leakage during rapid gas generation phases.

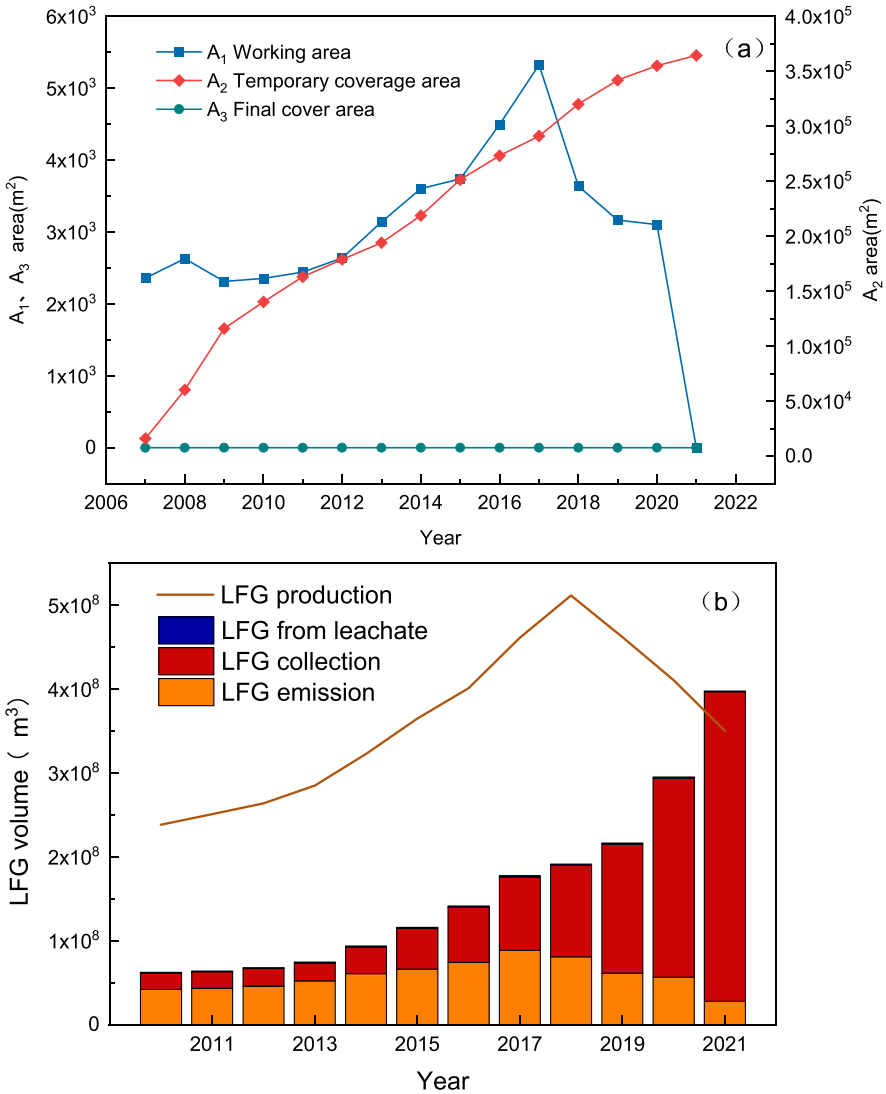

Fig. 5 Verification of LFG emissions**: a**) Temporal area changes in working area, temporary cover area, and final cover area during 2007–2021; **b**) Comparison of LFG production and various LFG emission sources during 2010–2021

The findings of this study provide valuable insights for China's landfill management policies and carbon neutrality goals: (1) Integration with China's National Carbon Trading System by establishing landfill-specific carbon credit mechanisms; (2) Alignment with the "14th Five-Year Plan" for waste management through setting specific targets for improving LFG collection rates nationwide; (3) Contributing to the 2060 carbon neutrality commitment by demonstrating that well-managed landfills can achieve net-negative emissions; (4) Informing policy development for mandatory LFG collection standards and incentive structures for landfill operators.

International comparison shows that HSL's performance aligns with EU landfills achieving > 75% LFG capture efficiency. However, China faces specific challenges including high organic waste content (60% vs. 30–40% in developed countries), leading to rapid gas generation phases requiring enhanced early collection. The framework's scalability to smaller landfills involves simplified monitoring systems and adapted collection technologies. For developing regions with limited technical resources, a phased implementation approach focusing on critical emission sources (uncollected LFG) provides the most cost-effective GHG reduction pathway.

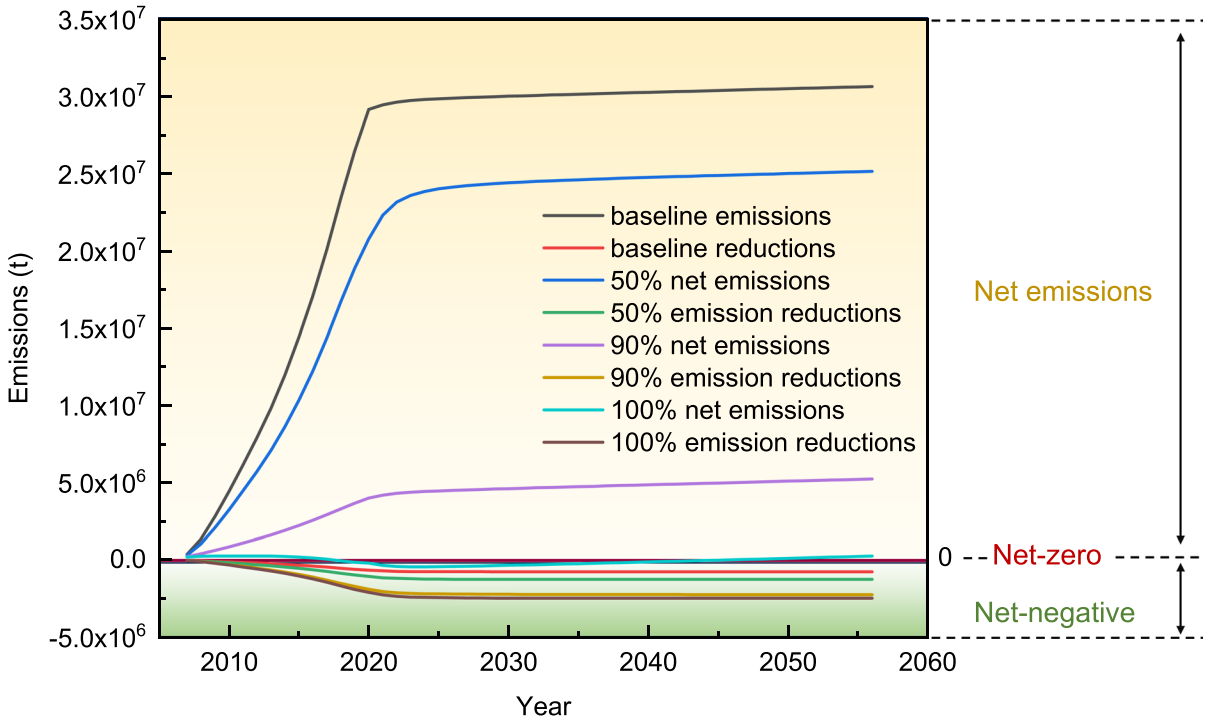

Fig. 6 Cumulative net GHG emissions and reductions in different collection rate situations (50%, 90%, and 100%)

6 Conclusions

Sanitary landfill is still a widely used waste treatment method worldwide, with high potential and value for reducing GHG emissions. Based on the material transformation law and energy conversion processes of MSW in sanitary landfills, a framework is proposed to account for the life-cycle GHG emissions and applied to the HSL. The possibility of achieving carbon neutrality for well-managed sanitary landfills is discussed. The main conclusions are as follows:

(1) GHG emissions in sanitary landfills primarily stem from the carbon and nitrogen present in MSW, as well as energy conversion processes. The carbon and nitrogen are released in the form of CH_4, CO_2, and N_2O. CH_4 emissions are mainly associated with the effective collection and utilization of LFG, whereas N_2O emissions primarily occur during leachate treatment and the collection and transportation of MSW. Energy conversion processes contributing to GHG emissions include fossil fuel combustion and electricity consumption.

(2) The proposed framework for accounting life-cycle GHG emissions in sanitary landfills is applied to a large-scale Hangzhou sanitary landfill. In the HSL study case, approximately 20.5 million tons of MSW are landfilled, resulting in the production of 6.33 million tons of LFG, 0.28 million tons of leachate, and 31.88 million tons of GHG emissions. GHG emissions from landfill buildings, collection and transportation and leachate treatment account for 2.95%, 4.35% and 2.9% of total GHG emissions, respectively. CH_4 from uncollected LFG contribute the most to GHG emissions, accounting for 89.8% of the total emissions.

(3) The LFG collection rate is the most crucial factor affecting GHG emissions from sanitary landfills. Increasing the LFG collection rate can reduce the GHG emissions of escaping CH_4, and if utilized for electricity generation, it can contribute to additional GHG reductions. Comparing three scenarios of LFG collection rates (50%, 90%, and 100%) to the baseline condition, a well-managed sanitary landfill has the potential to achieve carbon neutrality or even achieve negative net GHG emissions.

Acknowledgements This study was funded by the National Natural Science Foundation of China (Grant No.52588202; Grant No.52278376) and the Sichuan Provincial Regional Innovation Cooperation Program (Grant No.2024YFHZ0172.

Authors Contribution **Shuai Zhang**: Writing – review, Resources, Methodology, Investigation, Formal analysis, Funding acquisition. **Kai Liu**: Writing – original draft, Visualization, Validation. **Xuyang Huang**: Writing – review & editing. **He Li**: Writing – editing, Formal analysis. **Wenjie Xu**: Writing – review & editing, Data curation, Conceptualization, Supervision. **Yuzhong Zhang**: Writing – review, Formal analysis. **Guangyao Li**: Writing – review & editing, Formal analysis. **Liangtong Zhan**: Supervision, Resources, Project administration. **Jiwu Lan:** Investigation, Formal analysis, Resources, Project administration**. Yunmin Chen**: Supervision, Resources, Project administration.

Data Availability Data will be made available on request.

Declarations

Competing Interest The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

Anshassi, M., Sackles, H., & Townsend, T. G. (2021). A review of LCA assumptions impacting whether landfilling or incineration results in less greenhouse gas emissions. *Resources, Conservation and Recycling, 174*, Article 105810. https://doi.org/10.1016/j.resconrec.2021.105810

Arena, U., Mastellone, M. L., & Perugini, F. (2003). The environmental performance of alternative solid waste management options: A life cycle assessment study. *Chemical Engineering Journal, 2003*(96(1–3)), 207–222.

Atabi F, Ehyaei MA, Ahmadi MH (2014) Calculation of CH4 and CO2 emission rate in Kahrizak landfill site with LandGEM mathematical model. The 4th world sustainability forum. 2014, 18, https://doi.org/10.3390/wsf-4-e006

Bataille, C., Ahman, M., Neuhoff, K., Nilsson, L. J., Fischedick, M., Lechtenböhmer, S., Solano-Rodriguez, B., Denis-Ryan, A., Stiebert, S., & Waisman, H. (2018). A review of technology and policy deep decarbonization pathway options for making energy-intensive industry production consistent with the Paris Agreement. *Journal of Cleaner Production, 187*, 960–973. https://doi.org/10.1016/j.jclepro.2018.03.107

Brunner, P. H., & Rechberger, H. (2015). Waste to energy–key element for sustainable waste management. *Waste Management, 37*, 3–12. https://doi.org/10.1016/j.wasman.2014.02.003

Chen, Y., Guo, R., Li, Y. C., Liu, H., & Zhan, T. L. (2016). A degradation model for high kitchen waste content municipal solid waste. *Waste Management, 60*, 277–289. https://doi.org/10.1016/j.wasman.2016.08.011

Consonni, S., Giugliano, M., & Grosso, M. (2005). Alternative strategies for energy recovery from municipal solid waste: Part B: Emission and cost estimates. *Waste Management, 25*, 137–148. https://doi.org/10.1016/j.wasman.2004.09.006

De Feo, G., & Malvano, C. (2009). The use of LCA in selecting the best MSW management system. *Waste Management, 29*, 1901–1915. https://doi.org/10.1016/j.wasman.2008.12.021

Dong, H., Liu, Y., Zhao, Z., Tan, X., & Managi, S. (2022). Carbon neutrality commitment for China: From vision to action. *Sustainability Science, 17*, 1741–1755.

Eggleston, H., Buendia, L., Miwa, K., Ngara, T., & Tanabe, K. (2006). *2006 IPCC guidelines for national greenhouse gas inventories*. IPCC.

Fallahizadeh, S., Rahmatinia, M., Mohammadi, Z., Vaezzadeh, M., Tajamiri, A., & Soleimani, H. (2019). Estimation of methane gas by LandGEM model from Yasuj municipal solid waste landfill, Iran. *MethodsX, 6*, 391–398. https://doi.org/10.1016/j.mex.2019.02.013

Friedrich, E., & Trois, C. (2016). Current and future greenhouse gas (GHG) emissions from the management of municipal solid waste in the eThekwini Municipality-South Africa. *Journal of Cleaner Production, 112*, 4071–4083. https://doi.org/10.1016/j.jclepro.2015.05.118

Ghosh, P., Shah, G., Chandra, R., Sahota, S., Kumar, H., Vijay, V. K., & Thakur, I. S. (2019). Assessment of methane emissions and energy recovery potential from the municipal solid waste landfills of Delhi, India. *Bioresource Technology, 272*, 611–615. https://doi.org/10.1016/j.biortech.2018.10.069

He, H. J., & Hu, J. (2022). Leachate drainage volume of municipal solid waste landfills: Field testing and hydromechanical modeling. *Environmental Science and Pollution Research, 29*(43), 64680–64691. https://doi.org/10.1007/s11356-022-20413-9

IPCC (2019). 2019 Refinement to the 2006 IPCC Guidelines for National Greenhouse Gas Inventories. IPCC, Geneva, Switzerland.

Kaza, S., Yao, L. C., Bhada-Tata, P., & Woerden, F. V. (2018). *What a waste 2.0: A global snapshot of solid waste management to 2050*. World Bank Publications.

Liu, Y., Ni, Z., Kong, X., & Liu, J. (2017). Greenhouse gas emissions from municipal solid waste with a high organic fraction under different management scenarios. *Journal of Cleaner Production, 147*, 451–457. https://doi.org/10.1016/j.jclepro.2017.01.135

Manfredi, S., & Christensen, T. H. (2009). Environmental assessment of solid waste landfilling technologies by means of LCA-modeling. Waste management, 29(1), 32-43.

Pan, S. Y., Du, M. A., Huang, I. T., Liu, I. H., Chang, E., & Chiang, P. C. (2015). Strategies on implementation of waste-to-energy (WTE) supply chain for circular economy system: A review. *Journal of Cleaner Production, 108*, 409–421. https://doi.org/10.1016/j.jclepro.2015.06.124

Pandey, R. U., Surjan, A., & Kapshe, M. (2018). Exploring linkages between sustainable consumption and prevailing green practices in reuse and recycling of household waste: Case of Bhopal city in India. *Journal of Cleaner Production, 173*, 49–59. https://doi.org/10.1016/j.jclepro.2017.03.227

Pietzsch, N., Ribeiro, J. L. D., & de Medeiros, J. F. (2017). Benefits, challenges and critical factors of success for zero waste: A systematic literature review. *Waste Management, 67*, 324–353. https://doi.org/10.1016/j.wasman.2017.05.004

Ritchie H, Roser M, Rosado P (2020) CO_2 and greenhouse gas emissions. Our World in Data

Samadder, S. R., Prabhakar, R., Khan, D., Kishan, D., & Chauhan, M. S. (2017). Analysis of the contaminants released from municipal solid waste landfill site: A case study. *Science of the Total Environment, 580*, 593–601. https://doi.org/10.1016/j.scitotenv.2016.12.003

Tae, S., Shin, S., Woo, J., & Roh, S. (2011). The development of apartment house life cycle CO2 simple assessment system using standard apartment houses of South Korea. *Renewable and Sustainable Energy Reviews, 15*, 1454–1467. https://doi.org/10.1016/j.rser.2010.09.053

Turner, D. A., Williams, I. D., & Kemp, S. (2015). Greenhouse gas emission factors for recycling of source-segregated waste materials. *Resources, Conservation and Recycling, 105*, 186–197. https://doi.org/10.1016/j.resconrec.2015.10.026

U.S. Environmental Protection Agency (EPA), 2020. Advancing Sustainable Materials Management: Facts and Figures. EPA, Washington, DC.

Vaverková, M. D. (2019). Landfill impacts on the environment—review. *Geosciences, 9*(10), 431. https://doi.org/10.3390/geosciences9100431

Wang, Z., & Geng, L. (2015). Carbon emissions calculation from municipal solid waste and the influencing factors analysis in China. *Journal of Cleaner Production, 104*, 177–184. https://doi.org/10.1016/j.jclepro.2015.05.062

Wang, X., Jia, M., Zhang, C., Chen, S., & Cai, Z. (2017). Leachate treatment in landfills is a significant N2O source. *Science of the Total Environment, 596*, 18–25. https://doi.org/10.1016/j.scitotenv.2017.04.029

World Bank, (2020). Silpa, Kaza Lisa C., Yao Perinaz, Bhada-Tata Frank, Van Woerden What a Waste 2.0: A Global Snapshot of Solid Waste Management to 2050 Washington DC: World Bank

Xin, C., Zhang, T., Tsai, S. B., Zhai, Y. M., & Wang, J. (2020). An empirical study on greenhouse gas emission calculations under different municipal solid waste management strategies. *Applied Sciences, 10*, 1673. https://doi.org/10.3390/app10051673

Zhan, L. T., Xu, H., Chen, Y. M., Lü, F., Lan, J. W., Shao, L. M., Lin, W. A., & He, P. J. (2017). Biochemical, hydrological and mechanical behaviors of high food waste content MSW landfill: Preliminary findings from a large-scale experiment. *Waste Management, 63*, 27–40. https://doi.org/10.1016/j.wasman.2017.06.023

Zhang, X., & Wang, F. (2015). Life-cycle assessment and control measures for carbon emissions of typical buildings in China. *Building and Environment, 86*, 89–97. https://doi.org/10.1016/j.buildenv.2015.01.003

Zhao, W., Huppes, G., & Van der Voet, E. (2011). Eco-efficiency for greenhouse gas emissions mitigation of municipal solid waste management: A case study of Tianjin, China. *Waste Management, 31*, 1407–1415. https://doi.org/10.1016/j.wasman.2011.01.013

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.